Reg No.:_____ Name:_____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S2 (R) Exam May 2025 (2024 Scheme)

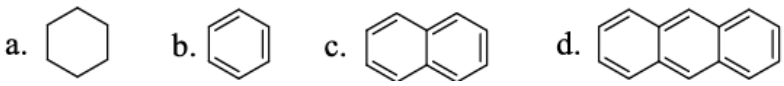
Course Code: GXCYT122

Course Name: CHEMISTRY FOR INFORMATION SCIENCE / ELECTRICAL SCIENCE

Max. Marks: 60 Duration: 2 hours 30 minutes

PART A

(Answer all questions. Each question carries 3 marks)

		CO	Marks
1	Write the anode, cathode reactions and calculate the standard EMF of the cell formed by silver and aluminium electrodes. Given that $E^0_{Ag^+/Ag} = 0.8V$ and $E^0_{Al^{3+}/Al} = -1.66V$.	CO1	(3)
2	Explain single electrode potential.	CO1	(3)
3	What are conducting conducting polymers? Give two examples.	CO2	(3)
4	Explain the sol-gel method for synthesis of nanomaterials.	CO2	(3)
5	How IR spectroscopy can be used for distinguishing intra and inter molecular hydrogen bonds?	CO3	(3)
6	Predict which of the following molecules will have highest λ_{max} in UV-visible spectrum. Justify your reason. 	CO3	(3)
7	Define dissolved oxygen. Give one significance of dissolved oxygen.	CO4	(3)
8	Explain water disinfection by chlorination.	CO4	(3)

PART B

(Answer any one full question from each module, each question carries 9 marks)

Module -1

9	a)	Explain the construction and working of Li-ion cell.	CO1	(6)
	b)	Explain impressed current cathodic protection.	CO1	(3)
10	a)	Discuss the construction and working of H_2-O_2 fuel cell using acid electrolyte. Give one advantage of H_2-O_2 fuel cell.	CO1	(6)

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	b)	Define reference electrodes. Give one example for primary and secondary reference electrode.	CO1	(3)

Module -2

11	a)	Discuss the construction and working of OLED. Give any two advantages of OLED.	CO2	(6)
	b)	Give any three applications of fullerenes.	CO2	(3)
12	a)	Describe the synthesis of polyaniline. List any two properties and applications of polyaniline.	CO2	(6)
	b)	Write any three materials used in spintronics.	CO2	(3)

Module -3

13	a)	Explain the various modes of vibrations possible for CO_2 and H_2O and state which of these modes are IR active.	CO3	(6)
	b)	List any three applications of SEM.	CO3	(3)
14	a)	Discuss the instrumentation and working of UV-Visible spectrometer.	CO3	(6)
	b)	Calculate the concentration of a solution if it shows a transmittance of 20% when taken in a cell of 2 cm thickness (Molar absorption coefficient is 12000 $dm^2\ mol^{-1}$).	CO3	(3)

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Module -4

15	a)	What is Greenhouse effect. Name any two greenhouse gases.	CO4	(3)
	b)	Explain trickling filter method of sewage treatment	CO4	(3)
	c)	List any three disadvantages of hard water.	CO4	(3)
16	a)	Explain the principle and procedure of ion exchange process of water softening.	CO4	(6)
	b)	Calculate the temporary and permanent hardness of water sample containing the following dissolved salts. $Mg(HCO_3)_2$ = 8 mg/L; $Ca(HCO_3)_2$ = 6 mg/L; $CaSO_4$ = 8 mg/L, MgSO4 = 10 mg/L	CO4	(3)

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